PhD Prospectus

Advancing Basin-Scale Groundwater Storage Estimation Through Hybrid Spatiotemporal Imputation, Empirical Orthogonal Function (EOF)-Based Interpolation, and Satellite-Derived Leakage Correction

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1. Introduction

1.1 Overall Objective and Significance

Groundwater sustains roughly half of global irrigation and a third of municipal water supply, yet it remains the most poorly monitored component of the terrestrial water budget. Jasechko et al. (2024) analyzed 170,000 monitoring wells across 1,693 aquifer systems on six continents and found that 30% exhibit accelerating decline. Reliable basin-scale quantification of groundwater storage change is essential for water-resources planning, yet three methodological barriers stand in the way: (1) temporal gaps in in situ well records that prevent continuous storage estimation; (2) the difficulty of converting sparse point observations into continuous spatial fields; and (3) the coarse resolution of satellite gravimetry (GRACE/GRACE-FO) which introduces signal leakage that must be corrected before groundwater storage can be isolated.

These three barriers form a chain: incomplete well records limit the quality of spatial fields, which in turn limit the fidelity of satellite-derived corrections. This dissertation addresses all three barriers in sequence through three papers, each building on the outputs of the previous:

1. **Paper 1** (submitted to journal) establishes baseline groundwater storage estimates for the Great Salt Lake Basin using multiple independent methods, revealing the sensitivity of GRACE-derived estimates to leakage handling and motivating the need for improved imputation.
2. **Paper 2** (in development) presents a novel hybrid imputation framework coupling Matrix Completion with Liquid Neural Networks (MC+LNN) that jointly exploits spatial cross-well correlations and continuous-time temporal dynamics -- a coupling that remains largely unexplored in groundwater imputation literature.
3. **Paper 3** applies the imputed records from Paper 2 to produce spatially continuous groundwater fields via Empirical Orthogonal Function (EOF)-based interpolation, and uses those fields to derive a pixel-wise GRACE leakage correction grid calibrated against spatially complete in situ data.

The Great Salt Lake Basin (GSLB) serves as the initial validation site, chosen for its dense USGS monitoring network (592 eligible wells), heterogeneous hydrogeology, and concentrated anthropogenic pumping. The methods are designed to be general-purpose, using globally available GLDAS auxiliary data with auto-optimized hyperparameters. Validation on additional basins -- including sites in Sub-Saharan Africa and South America -- is planned to demonstrate transferability.

1.2 Specific Objectives

Multi-Method Groundwater Storage Estimation (Paper 1). Quantify multi-decadal groundwater storage change in the GSLB (2002-2024) by integrating GRACE-derived, GLDAS-derived, and in situ estimates within a unified framework. Identify the methodological sensitivity to surface-water inclusion and GRACE leakage correction that motivates the spatially distributed approach of Paper 3. *(Submitted to journal.)*

Hybrid Spatiotemporal Imputation (Paper 2). Develop and validate a general-purpose hybrid imputation framework for sparse, irregular groundwater-level time series that couples PCHIP for short gaps with Matrix Completion + Liquid Neural Networks (MC+LNN) for long gaps, using globally available auxiliary climatic forcings.

EOF Interpolation and Pixel-Wise Leakage Correction (Paper 3). Apply the imputation framework from Paper 2 to produce spatially continuous groundwater-level fields via spatial trend decomposition and EOF analysis, and use those fields to derive a pixel-wise GRACE leakage correction grid calibrated against spatially complete in situ data.

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2. Multi-Method Groundwater Storage Estimation (Paper 1)

2.1 Objective

To quantify groundwater storage change in the Great Salt Lake Basin from 2002 through 2024 using multiple independent estimation methods, evaluate methodological sensitivities, and establish the baseline that motivates the imputation and leakage-correction advances of Papers 2 and 3.

2.2 Background

Three independent lines of evidence bear on basin-scale groundwater storage change. Satellite gravimetry from GRACE/GRACE-FO resolves total water storage anomalies at monthly resolution (Tapley et al., 2004), but the native spatial resolution of approximately 300 km makes basin-scale interpretation subject to signal leakage (Vishwakarma et al., 2018), and isolating the groundwater component requires subtracting auxiliary estimates from land-surface models (Rodell et al., 2004). In situ monitoring well networks provide direct observations but are temporally sparse.

The GSLB is a closed hydrologic system covering approximately 93,000 km-squared, chosen as the initial validation site for its dense USGS monitoring network, heterogeneous hydrogeology, and concentrated anthropogenic stress along the Wasatch Front.

Evans et al. (2020a) developed the Groundwater Data Mapper (GWDM), coupling Extreme Learning Machines (ELM; Huang et al., 2006) with Earth observation data to impute missing groundwater levels. Evans et al. (2020b) demonstrated that ELM with satellite-derived inputs outperforms conventional interpolation. Ramirez et al. (2022; 2023) extended this through inductive bias and iterative spatial-temporal refinement. Stevens et al. (2025) and Shepard et al. (2025) applied the GWDM methodology to the Central Valley and Klamath watershed, respectively.

2.3 Methods

Five GWSa estimates were computed: (1) GRACE-raw (TWSa minus GLDAS stores); (2) GRACE-sw (surface-water adjusted); (3) GRACE-Lf (leakage-corrected, Lf=2 calibrated against in situ data); (4) GLDAS-2.2 (GRACE-assimilated CLSM); and (5) GWDM (in situ via the GWDM workflow of Evans et al., 2020a, using PCHIP, ELM with Earth observation inputs, and kriging with specific yield 0.15). The five estimates were benchmarked against the independent GPS-based estimate of Young, Kreemer, and Blewitt (2021).

2.4 Results

All methods identified two major drawdowns (2012-2016, 2019-2022). The leakage-corrected GRACE estimate yielded 10.1 km-cubed loss for 2011-2016, consistent with the GPS estimate of 10.9 +/- 2.8 km-cubed (Young, Kreemer, and Blewitt, 2021). Surface water accounted for 31% of basin storage change. The basin-uniform leakage factor improved GRACE-in situ correlation from 0.17 to 0.77, but this uniform treatment masks known spatial heterogeneity in pumping and recharge -- motivating the pixel-wise approach of Paper 3. The in situ estimate was itself limited by well-record gaps and reliance on a single imputation method (ELM) -- motivating the improved imputation of Paper 2.

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3. Hybrid Spatiotemporal Imputation via Matrix Completion and Liquid Neural Networks (Paper 2)

3.1 Objective

To develop, validate, and benchmark a general-purpose hybrid imputation framework that jointly exploits spatial cross-well correlations (via matrix completion) and continuous-time temporal dynamics (via liquid neural networks) for reconstructing groundwater-level records with multi-year gaps.

3.2 Background

The ELM-based approaches described in Paper 1, while effective, operate primarily in the temporal domain: each well is imputed independently using auxiliary time series as features, with spatial information entering only through basin-average satellite-derived covariates rather than the actual cross-well correlation structure. Deep learning architectures face similar limitations. Jeong et al. (2020) demonstrated that recurrent neural networks achieve high accuracy but degrade sharply beyond one- to two-year gaps. Wunsch et al. (2021) compared LSTM, CNN, and NARX, finding multivariate models outperform univariate ones but remain limited by training data availability. Gharehbaghi et al. (2022) and Lin et al. (2022) showed GRU networks perform comparably to LSTM.

Regional correlation-based approaches represent an intermediate strategy. Levy et al. (2025) developed ARCHI, a USGS R package that imputes via iterative donor regression -- each well filled by linear regression from more complete reference wells. While ARCHI effectively exploits spatial correlation, it operates within a purely linear framework without auxiliary climatic forcings or nonlinear temporal modeling.

Matrix completion arranges all well records as a partially observed matrix and exploits low-rank structure to infer missing entries from cross-well correlations (Candes and Recht, 2009). Sharma, Kim, and Tayerani Charmchi (2024) evaluated SoftImpute for groundwater levels in the Chao-Phraya River Basin, finding it excels in sparse networks -- but treated it as a standalone method without temporal coupling.

On the temporal side, Liquid Neural Networks with Closed-form Continuous-depth cells (Hasani et al., 2022) model dynamics as continuous-time ODEs, naturally accommodating irregular sampling without resampling. To the authors' knowledge, matrix completion has not previously been coupled with continuous-time neural networks for groundwater imputation. The MC+LNN framework proposed here addresses this gap: MC provides spatially informed estimates from cross-well correlations, LNN refines using continuous-time dynamics conditioned on auxiliary forcings.

3.3 Methods

The framework operates on monthly-aggregated well observations via a two-stage pipeline:

Stage 1: PCHIP Small-Gap Fill. Gaps of 24 months or shorter are filled using Piecewise Cubic Hermite Interpolating Polynomials, which preserve monotonicity and local shape without oscillation. This densifies the monitoring network, providing spatial coverage needed for reliable donor correlation in Stage 2.

Stage 2: MC+LNN Large-Gap Fill. Gaps exceeding 24 months are filled in three phases:

Phase 2a -- Donor Regression. For each target well, the top 15 most correlated donor wells are selected from the PCHIP-densified network. Per-donor OLS regression provides a trend-aware initialization, following the donor-correlation concept of ARCHI (Levy et al., 2025) but extending it into a matrix-completion framework.

Phase 2b -- Matrix Completion. A composite matrix is constructed with rows representing the target well, weighted donor wells, GLDAS auxiliary variables (soil moisture at five temporal scales), and seasonal encoding. SoftImpute (iterative truncated SVD with adaptive rank selection) fills the target row's gaps by exploiting low-rank structure. Variance-preserving bias correction ensures predictions match observed statistics.

Phase 2c -- LNN Temporal Refinement. A Liquid Neural Network with Closed-form Continuous-time cells (Hasani et al., 2022) refines the MC predictions. The MC output serves as reservoir input during gap periods, while the LNN readout is trained exclusively on real observations via ridge regression -- ensuring ground-truth fidelity while benefiting from MC's spatial context. Hyperparameters are auto-optimized per well via grid search with 3-model ensemble selection by Kling-Gupta Efficiency.

Validation. Cross-validation on the GSLB (592 wells, 288 months, 2000-2023) under random missing data (5-50%, 50 trials each) and consecutive year gaps (1-5 years, 20 trials each). Additional basins are planned for transferability assessment.

3.4 Anticipated Results

Scenario	KGE	R-squared	RMSE (ft)
5% random missing	0.837 +/- 0.085	0.771	2.53
20% random missing	0.853 +/- 0.051	0.787	2.64
50% random missing	0.847 +/- 0.035	0.788	2.74

Scenario	KGE	R-squared	RMSE (ft)
1-year consecutive gap	0.783 +/- 0.116	0.703	2.98
3-year consecutive gap	0.802 +/- 0.076	0.730	3.02
5-year consecutive gap	0.815 +/- 0.063	0.744	2.91

KGE remains above 0.84 across all random-missing rates and above 0.78 for all consecutive-gap lengths. The PCHIP small-gap stage is critical: replacing it with LNN-based filling degrades KGE by 10-19%, because PCHIP densification improves donor correlations for the MC stage. Cross-validation across low-variance, medium-variance, and high-variance wells confirms robust performance across the full variance spectrum. The framework imputes all 592 GSLB wells to temporal completeness, producing the input dataset for Paper 3.

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4. EOF-Based Spatial Interpolation and Pixel-Wise GRACE Leakage Correction (Paper 3)

4.1 Objective

To produce spatially continuous groundwater-level fields from the imputed records of Paper 2 via EOF-based interpolation, and to use those fields to derive a pixel-wise GRACE leakage correction grid calibrated against spatially complete in situ data.

4.2 Background

Converting imputed point-well records to continuous spatial fields requires interpolation. Geostatistical methods, particularly kriging, have been dominant but their stationarity assumptions are frequently violated in heterogeneous basins where water-table elevation spans hundreds to thousands of meters (Ahmadi et al., 2024; Li et al., 2025). EOF analysis decomposes spatiotemporal fields into temporal modes and spatial loadings, enabling interpolation by estimating a few smooth spatial scalars rather than hundreds of raw time values.

The GRACE partition equation for groundwater storage anomalies requires a leakage correction factor Lf conventionally applied as a basin-uniform scalar. Long et al. (2014) established forward modeling for leakage correction. Ma et al. (2024) demonstrated that sub-regional trends can diverge substantially from basin averages. Tripathi et al. (2022) showed basin-average corrections can be misleading. Li et al. (2024) estimated pixel-scale factors using in situ data. However, all prior pixel-scale approaches have been constrained by the incompleteness of the underlying well records -- the limitation that Paper 2 addresses.

4.3 Methods

Spatial Interpolation. The complete imputed dataset (592 wells, 288 months) is interpolated via three stages: (1) a degree-2 polynomial trend surface of temporal-mean WTE as a function of latitude, longitude, and elevation (R-squared = 0.957); (2) EOF decomposition of the detrended residuals via SVD into k temporal modes and spatial loadings; (3) IDW interpolation of the spatial loadings to grid cells. This guarantees temporal coherence because all timesteps share the same modal structure.

Leakage Correction. The interpolated fields are converted to volumetric storage anomalies using spatially varying specific yield and aggregated to the 0.5-degree GRACE grid. A pixel-wise leakage factor Lf(phi, lambda) is calibrated against the in situ-derived GWSa. For grid cells lacking sufficient well coverage, Lf is propagated via a covariate-aware model.

Validation. The resulting pixel-wise GWSa is validated against the in situ reconstruction at sub-basin scales and compared with the basin-uniform Lf approach of Paper 1 to quantify the improvement from spatially distributed correction.

4.4 Anticipated Results

Leave-one-out cross-validation yields interpolation RMSE of 32.3 ft, a 49% reduction versus IDW (62.9 ft) and 73% versus per-timestep kriging (121.9 ft). The pixel-wise Lf grid is expected to show large factors along the Wasatch Front (concentrated pumping) and near-unity values in the West Desert (diffuse stress), departing substantially from the basin-uniform Lf = 2 of Paper 1 and improving sub-basin agreement with the in situ reconstruction.

5. Timeline

Period	Activity
May 2026	Paper 1 submitted to journal; Paper 2 method development in progress, validation ongoing
Jun-Aug 2026	Paper 2 manuscript drafting; begin cross-basin validation (Africa, South America sites)
Sep 2026	Paper 2 submission
Sep-Nov 2026	Paper 3 gridded Lf calibration, covariate propagation, sensitivity analyses
Dec 2026	Paper 3 results, leakage correction validation, cross-basin interpolation testing
Jan-Feb 2027	Paper 3 manuscript drafting and submission
Mar 2027	Dissertation compilation
Apr 2027	Dissertation defense and graduation

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